

Exercise 2: Probability measures

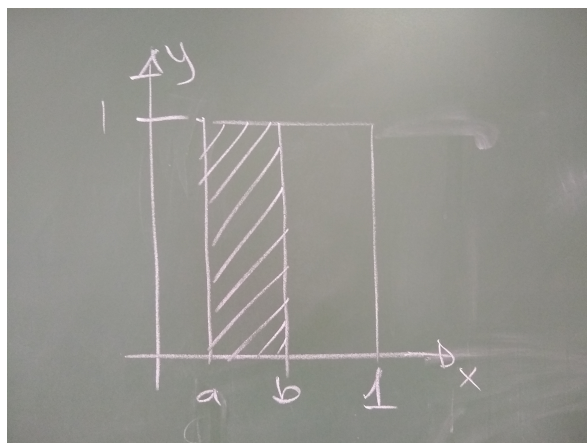
Consider the probability space $(\Omega, \mathcal{F}, \mathbf{P})$ with $\Omega = [0, 1]^2$, $\mathcal{F}$ the usual Borel-algebra on Ω , and $\mathbf{P}$ the Lebesgue measure, i.e. area. For $\omega = (x, y) \in \Omega$, define $X(\omega) = x$, $Y(\omega) = y$, and $Z(\omega) = x + y$.

Question 1 Independence and marginal distributions of X and Y : Verify that X and Y are independent, and that each are uniformly distributed on $[0, 1]$.

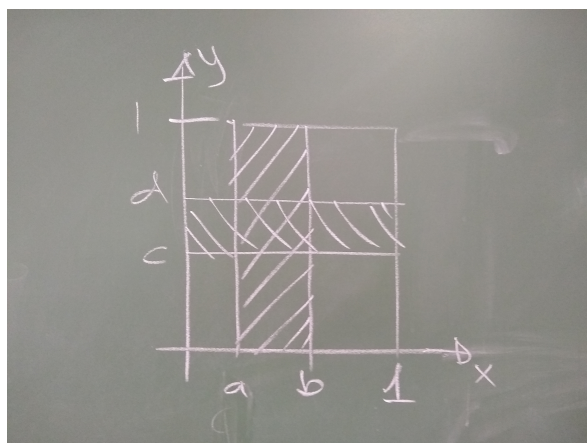
Solution: The following figure shows the sample space and the event $X \in [a, b]$. We see that the probability of this event is the area of the rectangle, i.e. $b - a$. With $a = 0$, we find

$$\mathbf{P}(X \leq b) = b$$

i.e., a uniform distribution.

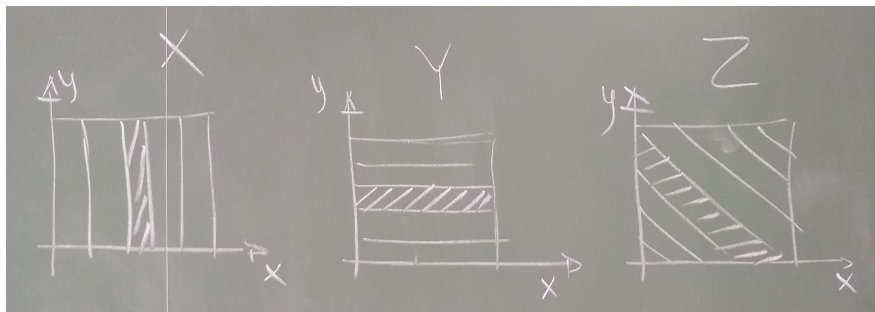


The next figure adds the event $Y \in [c, d]$. We see that $\mathbf{P}(X \in [a, b], Y \in [c, d]) = (b - a)(d - c) = \mathbf{P}(X \in [a, b])\mathbf{P}(Y \in [c, d])$, which defines independence. *Note:* Strictly, we should do this also for other events than intervals such as $X \in [a, b]$. This is a tedious step which we skip. It can be shown that since independence holds for intervals, and the Borel algebras $\sigma(X)$ and $\sigma(Y)$ are generated by intervals, independence holds for all such sets in $\mathcal{F}$: $\mathbf{P}(X \in A, Y \in B) = \mathbf{P}(X \in A)\mathbf{P}(Y \in B)$.



Question 2 Level sets and σ -algebras: Sketch level sets (contour lines) for X , Y , and Z .
Note: A level set for a function is a subset of the domain where the function attains one specific value, such as $X^{-1}(1/2) = \{(x, y) : x = 1/2\}$. Show typical elements in the σ -algebras $\sigma(X)$, $\sigma(Y)$ and $\sigma(Z)$.

Solution: The following figure shows typical level sets for X , Y and Z , along typical elements in the σ -algebras: Level sets for X are lines where $X(\omega) = x$ are constant, and typical elements in the σ -algebra are regions between the level sets (“cylinder sets”). Similarly for Y and Z .



Question 3 Monte Carlo simulation and conditional expectations of Z given X : Simulate a larger number, for example $N = 10,000$, of realizations of X , Y and Z . Binning the realized values of X into bins $[0, 1/n, 2/n, \dots, 1]$, estimate empirically the conditional expectations of Z given X

$$\mathbf{E} \left\{ Z | X \in \left[\frac{i-1}{n}, \frac{i}{n} \right] \right\}$$

for $n = 10$ and $i = 1, \dots, n$. Plot the result against i/n .

Solution: The following code does the simulation, the binning and the conditional expectations. We include also the analytical result from the next question.

```
N <- 1e4

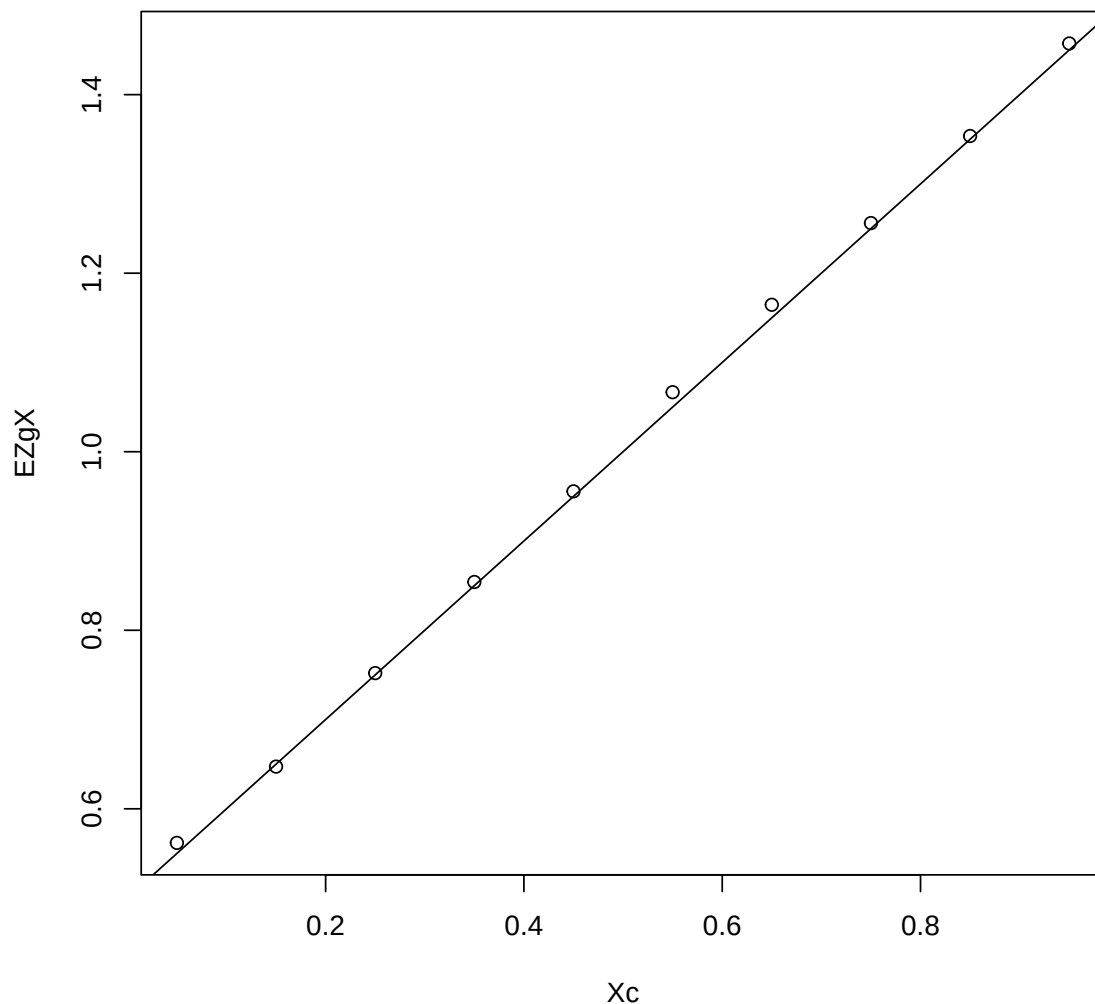
X <- runif(N)
Y <- runif(N)
Z <- X+Y

dat <- data.frame(X=X,Y=Y,Z=Z)

Xbreaks <- seq(0,1,0.1)
Xc <- Xbreaks[-1] - 0.5*diff(Xbreaks)

EZgX <- tapply(Z,cut(X,breaks=Xbreaks),mean)

plot(Xc,EZgX)
plot(function(x)x+0.5,add=TRUE,from=0,to=1)
```



Question 4 Conditional expectations of Z given X , analytically: Find a function $g : \mathbf{R} \mapsto \mathbf{R}$ such that $\mathbf{E}\{Z|X\} = g(X)$. Use inspiration (!) from the Monte Carlo experiments and elementary arguments. Sketch contour lines of $\mathbf{E}\{Z|X\}$. Verify that this “candidate” conditional expectation satisfies the defining property

$$\int_H g(X(\omega)) \, d\mathbf{P}(\omega) = \int_H Z(\omega) \, d\mathbf{P}(\omega)$$

for a typical $H \in \sigma(X)$.

Solution: We get:

$$\begin{aligned} \mathbf{E}\{Z|X\} &= \mathbf{E}\{X + Y|X\} \quad (\text{definition of } Z) \\ &= \mathbf{E}\{X|X\} + \mathbf{E}\{Y|X\} \quad (\text{linearity of conditional expectations}) \\ &= X + \mathbf{E}\{Y\} \quad (X \text{ is } X\text{-measurable; } Y \text{ and } X \text{ are independent}) \\ &= X + \frac{1}{2} \quad (Y \text{ is uniformly distributed}) \end{aligned}$$

so $g(x) = x + \frac{1}{2}$.

The contour lines of $\mathbf{E}\{Z|X\}$ agree with those of X , but the levels are different (the difference being $\frac{1}{2}$).

To verify the definition of conditional expectation, we take H to be the cylinder set

$$H = \{\omega = (x, y) : a \leq x \leq b, 0 \leq y \leq 1\}.$$

We get

$$\int_H g(X(\omega)) \, d\mathbf{P}(\omega) = \int_a^b \int_0^1 x + \frac{1}{2} \, dy \, dx = \frac{1}{2}(b^2 - a^2) + \frac{b-a}{2}.$$

Meanwhile, we get

$$\int_H Z(\omega) \, d\mathbf{P}(\omega) = \int_a^b \int_0^1 x + y \, dy \, dx = \frac{1}{2}(b^2 - a^2) + \frac{b-a}{2}$$

which verifies the definition. Again, it can be shown that since this holds for any event $H = X^{-1}([a, b])$, it holds for any X -measurable set.

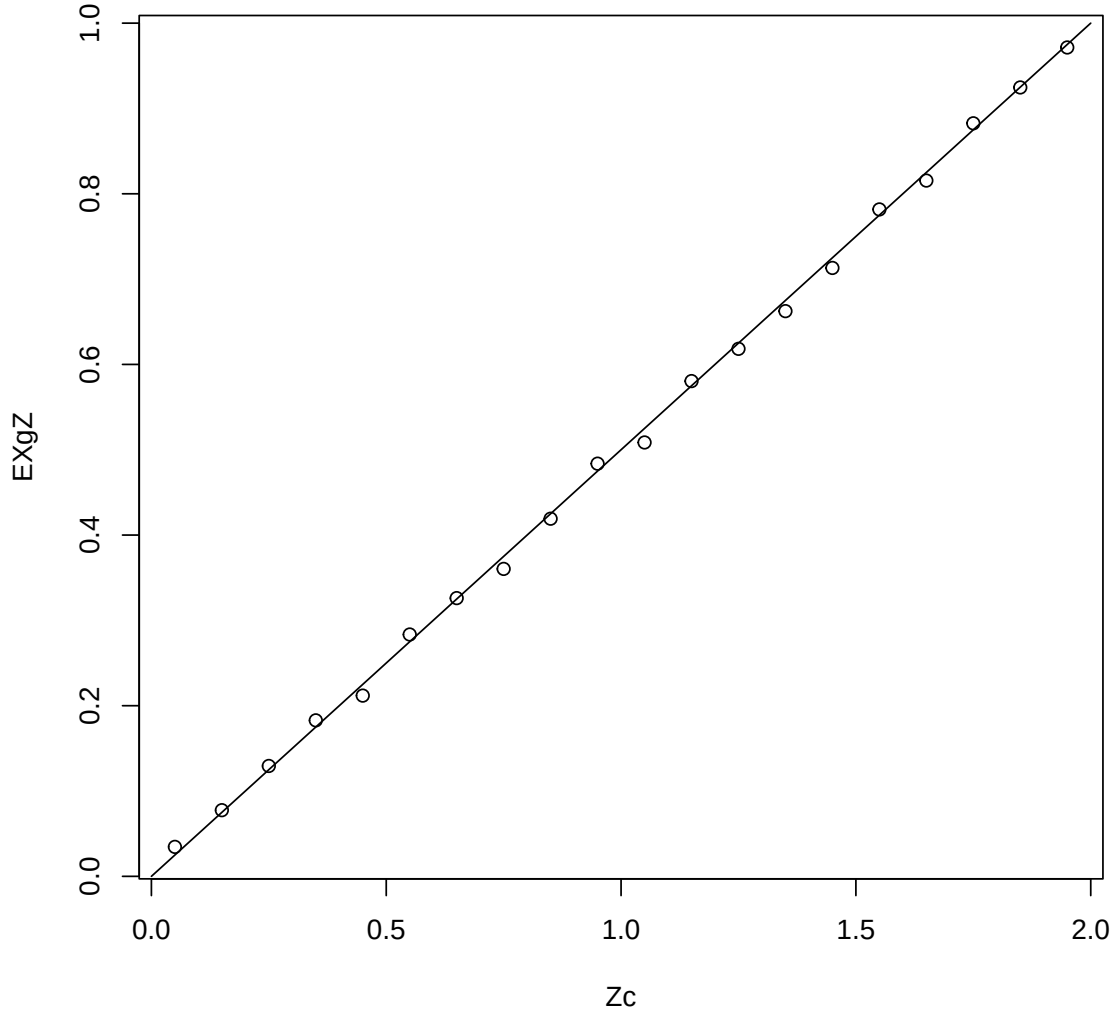
Question 5 Conditional expectations of X given Z : Repeat the two previous questions for $\mathbf{E}\{X|Z\}$: First estimate the function $h : \mathbf{R} \mapsto \mathbf{R}$ such that $h(Z) = \mathbf{E}\{X|Z\}$ from the Monte Carlo experiments. Then find that function and verify that it satisfies its defining properties.

Solution: The following code does the simulation. We include the analytical result $\mathbf{E}\{X|Z\}$, which we find in the following.

```
Zbreaks <- seq(0,2,0.1)
Zc <- Zbreaks[-1] - 0.5*diff(Zbreaks)

cZ <- cut(Z,breaks=Zbreaks)

EXgZ <- tapply(X,cZ,mean)
plot(Zc,EXgZ)
plot(function(z)0.5*z,add=TRUE,from=0,to=2)
```



The result $\mathbf{E}\{X|Z\} = \frac{1}{2}Z$ is most easily obtained with an appeal to symmetry: We must have

$$\mathbf{E}\{X|Z\} + \mathbf{E}\{Y|Z\} = \mathbf{E}\{Z|Z\} = Z$$

and

$$\mathbf{E}\{X|Z\} = \mathbf{E}\{Y|Z\}$$

Thus $h(z) = z/2$.

To verify that this function satisfies the defining properties, we first define $\hat{X} = \mathbf{E}\{X|Z\}$. We need a "typical element" $H \in \sigma(Z)$; we have previously sketched such an element. For simplicity, we consider a "thin" set where $Z \in [z, z + dz]$. There are two cases:

1. $z < 1$. Then the area of the set is $z \, dz$. We get $\hat{X} = z/2$ and $\mathbf{E}\{\hat{X}\mathbf{1}_H\} = z^2/2 \, dz$. For comparison, we get $\mathbf{E}\{X\mathbf{1}_H\} = \int_0^z x \, dx \, dz = z^2/2 \, dz$.
2. $z > 1$. Then the area of the set is $(2 - z) \, dz$. We get $\mathbf{E}\{\hat{X}|Z\} = \frac{z}{2}(2 - z) \, dz$. For comparison, we get $\mathbf{E}\{X\mathbf{1}_H\} = \int_{z-1}^1 x \, dx \, dz = \frac{1}{2}(1 - (z - 1)^2) \, dz = \frac{1}{2}(2z - z^2) \, dz = \frac{z}{2}(2 - z) \, dz$.

In either case, we see that $Z/2$ satisfies the defining property of $\mathbf{E}\{X|Z\}$.

Question 6 The law of total expectation (The simple Tower property): Verify from the Monte Carlo experiments that

$$\mathbf{E}\{\mathbf{E}\{X|Z\}\} = \mathbf{E}\{X\}$$

Solution: The verification of the can be done in two ways: We can use that $\hat{X} = Z/2$ and verify that $\mathbf{E}X = \mathbf{E}\hat{X}$:

```
## Based on analytical conditioning:
```

```
mean(X)
```

```
## [1] 0.5004579
```

```
mean(Z/2)
```

```
## [1] 0.5037824
```

Alternatively, we can do this empirically, where we condition not on true Z , but on the bin in which Z lies:

```
## Based on binned Z's:
```

```
mean(X)
```

```
## [1] 0.5004579
```

```
pZ <- table(cZ) / length(Z)
```

```
sum(EXgZ*pZ)
```

```
## [1] 0.5004579
```

In the first case, we get a slight discrepancy due to the random Y -values. In the second case, we get exact equality.

Question 7 Variance decomposition/Law of total variance: Find an analytical expression for $\mathbf{V}\{X|Z\}$ and verify it against the Monte Carlo simulations. Then verify from the Monte Carlo simulations the variance decomposition formula

$$\mathbf{V}\{X\} = \mathbf{E}\mathbf{V}\{X|Z\} + \mathbf{V}\mathbf{E}\{X|Z\}$$

Solution: If $Z < 1$, then conditional on Z , X is uniformly distributed on $[0, Z]$ and therefore has conditional variance $Z^2/12$. If $Z > 1$, then X is uniformly distributed on $[Z - 1, 1]$ and therefore has conditional variance $(2 - Z)^2/12$. The following code verifies this expression:

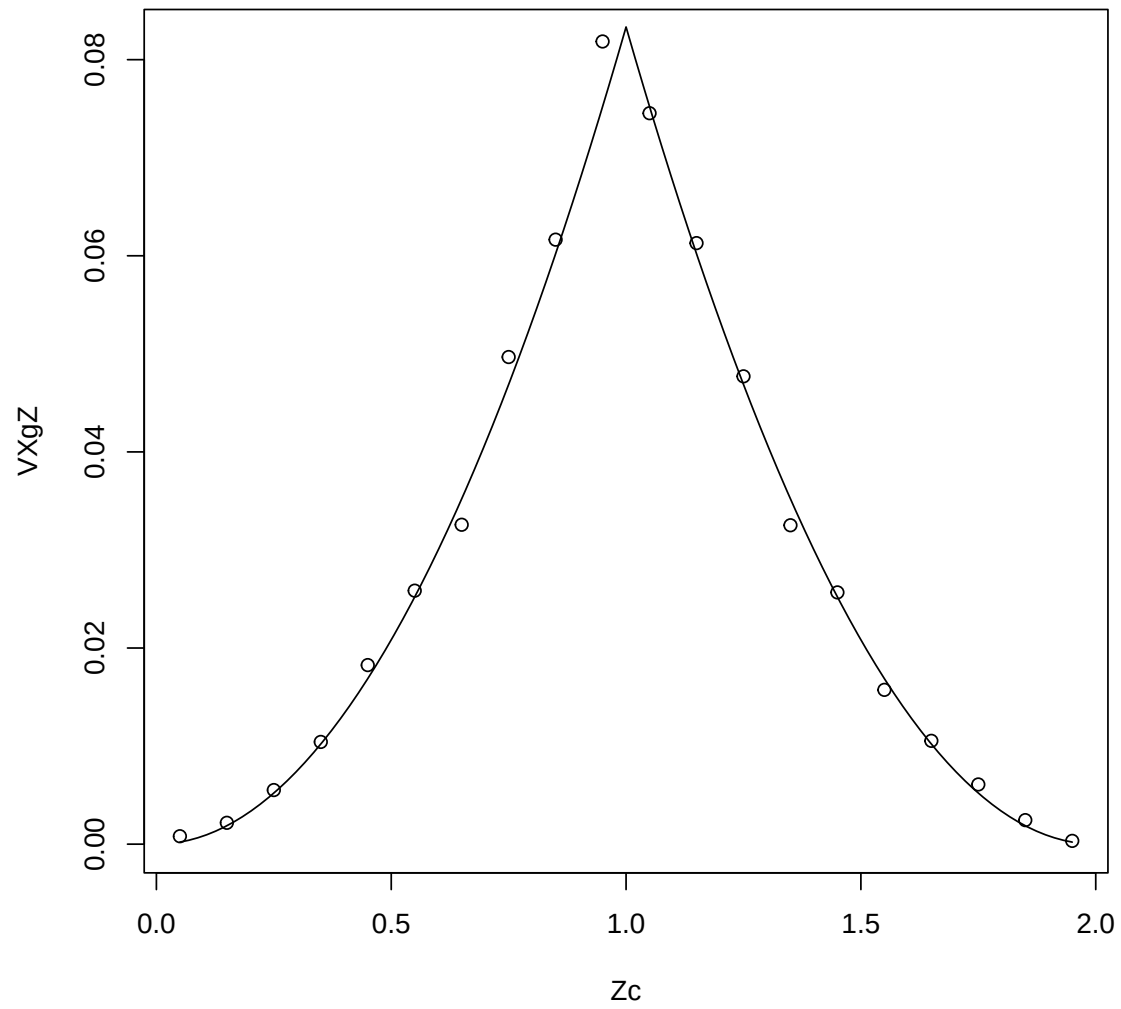
```
## Conditional variance
```

```
VXgZ <- tapply(X, cZ, var)
```

```
plot(Zc, VXgZ)
```

```
VXgzanalytical <- function(z) pmin(z, 2-z)^2/12
```

```
curve(VXgzanalytical, add=TRUE)
```



The verification can again be done purely empirical, or using the analytical expressions:

```

## Variance decomposition, analytically:
print(var(X))

## [1] 0.0836867

print(mean(VXgzanalytical(Z)) + var(Z/2))

## [1] 0.08340128

## Variance decomposition, purely empirical:
VXgZ[is.na(pZ)] <- 0
EXgZ[is.na(pZ)] <- 0
pZ[is.na(pZ)] <- 0

print(sum(pZ*VXgZ) + sum(pZ*(EXgZ-sum(pZ*EXgZ))^2))

## [1] 0.0837349

```